

CSS Grid - Complete Beginner Guide

CSS Grid is a powerful 2-dimensional layout system used to design web pages with rows and columns. It helps developers create responsive and flexible layouts easily.

1. What is CSS Grid?

CSS Grid allows you to divide a webpage into rows and columns and place elements exactly where you want.

2. Grid Container

To start using Grid, set `display: grid` on a parent container.

3. Grid Columns

Use `grid-template-columns` to define the number and size of columns.

4. Grid Rows

Use `grid-template-rows` to define rows.

5. Gap Property

The `gap` property adds spacing between rows and columns.

6. Fraction Unit (fr)

The `fr` unit divides available space proportionally.

7. Grid Item Placement

Use `grid-column` and `grid-row` to control item positions.

8. Repeat Function

`repeat()` helps avoid writing the same values multiple times.

9. Responsive Grid

Use `auto-fit`, `auto-fill`, and `minmax()` for responsive layouts.

10. Grid Areas

`grid-template-areas` creates named layout sections.

Important CSS Grid Examples

Basic Grid Container

```
container {  
  display: grid;  
  grid-template-columns: 200px 200px 200px;  
}
```

Using fr Units

```
container {  
  display: grid;  
  grid-template-columns: 1fr 2fr 1fr;  
}
```

Using Gap

```
container {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 20px;  
}
```

Grid Item Placement

```
item1 {  
  grid-column: 1 / 3;  
  grid-row: 1 / 2;  
}
```

Responsive Grid

```
container {  
  display: grid;  
  grid-template-columns:  
    repeat(auto-fit, minmax(200px, 1fr));  
}
```

Grid Areas

```
container {  
  display: grid;  
  grid-template-areas:  
    "header header"  
    "sidebar main"  
    "footer footer";  
}
```

Flexbox vs Grid

Feature	Flexbox	Grid
Dimension	1D Layout	2D Layout
Best For	Small layouts	Full page layouts
Direction	Row OR Column	Rows AND Columns
Complex Design	Limited	Excellent

Tips for Beginners

- Use Grid for complete page layouts.

- Use Flexbox for alignment inside components.
- Practice with small projects like galleries and dashboards.
- Learn both Grid and Flexbox together for modern web development.

CSS Grid is one of the most important modern CSS concepts. Mastering Grid will help you create professional and responsive web layouts easily.